

A CAUCHY LIMIT FOR A SAWTOOTH SUM

ABSTRACT. For $S_n(\alpha) = \sum_{k \leq n} (1/2 - \{k\alpha\})$, with α uniformly distributed on $(0, 1)$, we prove that $S_n(\alpha)/\log n$ converges in distribution to the centered Cauchy law of scale $1/(2\pi)$. An exact Euclidean-algorithm reciprocity formula reduces the sum to an alternating continued-fraction cost. Its principal part is a Gauss digit marked by a rapidly oscillating torus coordinate, while the remaining carry term is bounded. Local cylinder oscillation, a resonance decomposition, and a pullback carry construction give the required Poisson limit for large jumps and a weak law for the bounded remainder. Consequently the limiting distribution function is $\frac{1}{2} + \pi^{-1} \arctan(2\pi c)$.

AI-use disclosure. The proposed answer was first obtained in a one-shot ChatGPT response using OpenAI GPT-5.6-sol Pro. The proof was subsequently developed and gaps were addressed through iterative prompting of that model. OpenAI Codex (GPT-5.6-sol) was used to assemble, edit, compile, and perform automated consistency audits of this manuscript.

1. INTRODUCTION AND STATEMENT OF THE RESULT

Write $\{x\} = x - \lfloor x \rfloor$ and let logarithms be natural. For

$$S_n(\alpha) = \sum_{k=1}^n \left(\frac{1}{2} - \{k\alpha\} \right), \quad 0 < \alpha < 1,$$

the question is whether $S_n(\alpha)/\log n$, regarded as a random variable on the Lebesgue probability space $(0, 1)$, has a limiting law. This is Erdős Problem #1002, originating in [5]; see also [2]. Kesten's two-parameter result [6] provides the closely related averaged-origin case; Borda [3] computed the normalizing constant in that law for the sawtooth observable. This is an annealed spatial law, averaging both the rotation number and the starting point. Dolgopyat and Sarig [4] proved an annealed temporal Cauchy law in which the rotation number and the time are randomized. Neither ensemble is the one considered here: the time n and the starting point are fixed, and only α is averaged. The answer to this fixed-origin, fixed-time problem is as follows.

Theorem 1.1. *As $n \rightarrow \infty$,*

$$\frac{S_n(\alpha)}{\log n} \implies \text{Cauchy}\left(0, \frac{1}{2\pi}\right).$$

Equivalently, for every $c \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \text{Leb} \left\{ \alpha \in (0, 1) : \frac{S_n(\alpha)}{\log n} \leq c \right\} = \frac{1}{2} + \frac{1}{\pi} \arctan(2\pi c).$$

Here $\text{Cauchy}(0, \gamma)$ denotes the centered Cauchy law with characteristic function $t \mapsto \exp(-\gamma|t|)$.

The proof has four parts. First an exact reciprocity identity expresses S_n as an alternating sum along the Gauss map. Second, continued-fraction coordinates separate each summand into a heavy-tailed marked digit and a bounded carry remainder. Third, the marked digits satisfy a Poisson limit; the main point is that the initial measure lies on the oscillating graph $\alpha \mapsto (\alpha, \{n\alpha\})$, so ordinary Gauss mixing is insufficient. We instead integrate Fourier modes on complete continued-fraction cylinders, before and after their unique resonance scale. Finally, the carry is realized as a measurable graph over the natural extension of a Gauss–torus cocycle; a reset argument and a two-block correlation estimate give a weak law for the bounded remainder.

Throughout, constants denoted by C, c may change from line to line. A bound $O_A(L^{-A})$ is uniform in the time indices satisfying the displayed hypotheses. We put $L = \log n$ whenever n is fixed.

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2. EXACT RENORMALIZATION AND CONTINUED-FRACTION COORDINATES

Let $T : (0, 1) \rightarrow (0, 1)$ be the Gauss map,

$$Tx = \left\{ \frac{1}{x} \right\}, \quad a_1(x) = \left\lfloor \frac{1}{x} \right\rfloor.$$

Rational points will always be ignored, since they form a null set.

Proposition 2.1 (Reciprocity). *For irrational $x \in (0, 1)$ and $N \geq 1$, set*

$$m = \lfloor Nx \rfloor, \quad u = \{Nx\}.$$

Then

$$(1) \quad S_N(x) = -S_m(Tx) + \Phi(x, u), \quad \Phi(x, u) = \frac{u(1-u)}{2x} - \frac{u}{2}.$$

Proof. Since x is irrational,

$$S_N(x) = \frac{N}{2} - \frac{xN(N+1)}{2} + \sum_{k=1}^N \lfloor kx \rfloor.$$

Counting lattice points below $j = kx$ in the other order gives

$$\sum_{k=1}^N \lfloor kx \rfloor = Nm - \sum_{j=1}^m \left\lfloor \frac{j}{x} \right\rfloor.$$

If $a = \lfloor 1/x \rfloor$, then $1/x = a + Tx$, and hence

$$\sum_{j=1}^m \left\lfloor \frac{j}{x} \right\rfloor = \frac{am(m+1)}{2} + \sum_{j=1}^m \lfloor jTx \rfloor.$$

Substitution and the identity $m = Nx - u$ reduce the remaining polynomial expression to $u(1-u)/(2x) - u/2$. $\square$

Starting with $N_0 = n$, $x_0 = \alpha$, define

$$(2) \quad N_{j+1} = \lfloor N_j x_j \rfloor, \quad x_{j+1} = Tx_j, \quad u_j = \{N_j x_j\}.$$

Let $\tau_n = \min\{j : N_j = 0\}$. Iterating Proposition 2.1 gives the exact identity

$$(3) \quad S_n(\alpha) = \sum_{j=0}^{\tau_n-1} (-1)^j \Phi(x_j, u_j).$$

Write

$$\alpha = [0; a_1, a_2, \dots], \quad x_j = T^j \alpha, \quad a_{j+1} = a_1(x_j),$$

and let p_j/q_j be the convergents, normalized by

$$p_{-1} = 1, \quad p_0 = 0, \quad q_{-1} = 0, \quad q_0 = 1.$$

The standard identities

$$(4) \quad \beta_j := x_0 x_1 \cdots x_j = (-1)^j (q_j \alpha - p_j) = \frac{1}{q_{j+1} + q_j x_{j+1}}$$

will be used repeatedly; set $\beta_{-1} = 1$.

Define the continuous heights and their errors by

$$C_j = n\beta_{j-1}, \quad E_j = C_j - N_j.$$

Then $C_{j+1} = x_j C_j$, and (2) yields

$$(5) \quad E_{j+1} = x_j E_j + u_j, \quad E_0 = 0.$$

For every $r, \ell \geq 0$,

$$(6) \quad x_r x_{r+1} \cdots x_{r+\ell-1} \leq F_{\ell+1}^{-1},$$

where $F_1 = F_2 = 1$. Indeed, the reciprocal of the product is a continuant with all digits at least one. Iterating (5) therefore gives

$$(7) \quad 0 \leq E_j \leq E_* := \sum_{\ell \geq 1} F_\ell^{-1} < \infty$$

uniformly in j, n, α .

Put

$$(8) \quad \theta_j = \{n\beta_j\} = \{(-1)^j n q_j \alpha\}, \quad \theta_{-1} = 0.$$

From $q_{j+1} = a_{j+1}q_j + q_{j-1}$ we obtain

$$(9) \quad \theta_{j+1} = \{\theta_{j-1} - a_{j+1}\theta_j\}.$$

Since $\{E_j\} = \theta_{j-1}$, write

$$E_j = d_j + \theta_{j-1}, \quad d_j = \lfloor E_j \rfloor.$$

The equality $C_{j+1} = N_{j+1} + u_j + x_j E_j$ implies

$$(10) \quad u_j = \{\theta_j - x_j(d_j + \theta_{j-1})\}.$$

Let

$$W(t) = \frac{\{t\}(1 - \{t\})}{2}, \quad t \in \mathbb{R}/\mathbb{Z}.$$

This function is $1/2$ -Lipschitz and $0 \leq W \leq 1/8$.

Proposition 2.2 (Principal term). *There is an absolute constant C_0 such that, for every j ,*

$$(11) \quad \Phi(x_j, u_j) = a_{j+1}W(\theta_j) + B_j, \quad |B_j| \leq C_0.$$

Proof. By (10) and (7),

$$\text{dist}_{\mathbb{T}}(u_j, \theta_j) \leq x_j E_*.$$

Since $x_j^{-1} = a_{j+1} + x_{j+1}$,

$$\Phi(x_j, u_j) - a_{j+1}W(\theta_j) = a_{j+1}\{W(u_j) - W(\theta_j)\} + x_{j+1}W(u_j) - \frac{u_j}{2}.$$

The first term has absolute value at most $a_{j+1}x_j E_*/2 \leq E_*/2$, and the other two are bounded. $\square$

3. GAUSS ESTIMATES AND LOCAL OSCILLATION

The Gauss invariant probability measure is

$$(12) \quad d\nu(x) = h(x) dx, \quad h(x) = \frac{1}{\log 2 (1+x)}.$$

Its Lyapunov exponent is

$$(13) \quad \lambda = \int_0^1 -\log x d\nu(x) = \frac{\pi^2}{12 \log 2}.$$

For a word $w = (a_1, \dots, a_r)$ let I_w be the corresponding continued-fraction cylinder. If p_r/q_r is its last convergent, the inverse branch is

$$h_w(y) = \frac{p_r + p_{r-1}y}{q_r + q_{r-1}y}, \quad |h'_w(y)| = (q_r + q_{r-1}y)^{-2},$$

and

$$(14) \quad |I_w| = \frac{1}{q_r(q_r + q_{r-1})}.$$

Conditionally on I_w , the density of $T^r \alpha = y$ is

$$\rho_w(y) = \frac{q_r(q_r + q_{r-1})}{(q_r + q_{r-1}y)^2}.$$

Thus $\sup_w \|\rho_w\|_{BV} < \infty$.

We use the following standard consequences of the Lasota–Yorke inequality for the Gauss transfer operator; see, for example, [1]. They are recorded with the uniformity needed below.

Lemma 3.1 (Gauss estimates). *There are $C > 0$ and $0 < \rho < 1$ such that:*

(1) *for $f \in BV(0, 1)$,*

$$\|\mathcal{L}^r f - h \int f\|_1 \leq C \rho^r \|f\|_{BV};$$

(2) *conditionally on any past cylinder,*

$$\mathbb{P}(a_{j+1} \geq A \mid a_1, \dots, a_j) \leq C/A;$$

in particular, for distinct $j_1 < \dots < j_s$,

$$(15) \quad \mathbb{P}(a_{j_i+1} \geq A_i, \ 1 \leq i \leq s) \leq C^s \prod_{i=1}^s A_i^{-1};$$

(3) *uniformly for $r \geq 1$ and $v > 0$,*

$$(16) \quad \mathbb{P}(|\log q_r - \lambda r| > v) \leq C \exp\{-c \min(v^2/r, v)\}.$$

Proof. For completeness, the transfer operator is

$$\mathcal{L}f(x) = \sum_{a \geq 1} (a+x)^{-2} f((a+x)^{-1}).$$

On two-step inverse branches the contraction is at most $1/2$; the sums of the inverse Jacobians and of their variations converge. The branchwise chain rule gives

$$\text{Var}(\mathcal{L}^2 f) \leq \rho_0 \text{Var}(f) + C \|f\|_1, \quad \rho_0 < 1.$$

Helly compactness and exactness yield (i). The conditional densities ρ_w have uniformly bounded variation, and $\{a_1 \geq A\} = (0, 1/A]$ up to endpoints; this proves (ii), and iteration proves (15). Finally, the perturbed operators

$$\mathcal{L}_t f(x) = \sum_{a \geq 1} (a+x)^{-2+t} f((a+x)^{-1})$$

form an analytic family on BV for $|t| < 1/2$. Analytic perturbation of the simple leading eigenvalue gives an eigenvalue $\kappa(t)$ and a uniformly bounded spectral projector such that

$$\log \kappa(t) = \lambda t + O(t^2), \quad \mathbb{E} \exp\left(t \sum_{i < r} -\log T^i \alpha\right) \leq C \kappa(t)^r$$

for $|t|$ sufficiently small; the same estimate holds for negative t . For $0 < v \leq r$, Chernoff's inequality with $|t| \asymp v/r$ gives $C \exp(-cv^2/r)$, while for $v > r$ a fixed sufficiently small $|t|$ gives $C \exp(-cv)$. Thus the asserted Bernstein bound holds for $\sum_{i < r} -\log T^i \alpha$. Since

$$\left| \log q_r - \sum_{i < r} -\log T^i \alpha \right| \leq \log 2,$$

we obtain (iii). □

Lemma 3.2 (Conditional multi-block mixing). *Let I_w be a cylinder of depth d , and let $d + M \leq t_1 < \dots < t_s$ with $t_{i+1} - t_i \geq M$. If $g_1, \dots, g_s \in BV(0, 1)$, then*

$$(17) \quad \left| \mathbb{E} \left(\prod_{i=1}^s g_i(T^{t_i} \alpha) \mid I_w \right) - \prod_{i=1}^s \int g_i d\nu \right| \leq C_s \rho^M \prod_{i=1}^s \|g_i\|_{BV}.$$

The constants are uniform in the cylinder I_w .

Proof. Conditioned on I_w , the density of $T^d \alpha$ is ρ_w , whose BV norm is uniformly bounded by (14). Apply Lemma 3.1(i) across the first gap, multiply by the next BV observable, and iterate. The product inequality $\|fg\|_{BV} \leq C \|f\|_{BV} \|g\|_{BV}$ and the fixed number s of factors give (17). □

Fix henceforth

$$(18) \quad L = \log n, \quad H = L^{3/4}, \quad m_n = \lfloor L/\lambda \rfloor,$$

and define the bulk set

$$(19) \quad \mathcal{J}_n = \{j \in \mathbb{Z} : 200H \leq j \leq m_n - 200H\}.$$

By (16), for every fixed $\delta > 0$ and $t \leq 2m_n$,

$$(20) \quad \mathbb{P}(q_t \notin [e^{\lambda t - \delta H}, e^{\lambda t + \delta H}]) \leq C_\delta e^{-c_\delta L^{1/2}}.$$

We always use (20) as a union of complete depth- t cylinders. We never insert a global good-event indicator inside an oscillatory integral.

The first ingredient controls local cancellation of the two continuants in a torus character.

Lemma 3.3 (Shrinking anti-concentration). *For every fixed $(r, s) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$, every $j \geq 1$, and $0 < \eta < 1/2$,*

$$(21) \quad \mathbb{P}(|sq_j - rq_{j-1}| < \eta q_j) \leq C_{r,s}\eta + C_{r,s}e^{-cj}.$$

Proof. Put $Y_j = q_{j-1}/q_j = [0; a_j, \dots, a_1]$ and $Y_{j,R} = [0; a_j, \dots, a_{j-R+1}]$. The contraction of inverse continued fractions gives

$$|Y_j - Y_{j,R}| \leq C\varphi^{-2R}, \quad \varphi = (1 + \sqrt{5})/2.$$

The Gauss natural extension has invariant density

$$\frac{dx dy}{\log 2 (1 + xy)^2};$$

its backward marginal is $dy/(\log 2(1 + y))$. By Lemma 3.1(i), the law of the last R digits under Lebesgue measure differs in total variation from its stationary law by $O(\rho^{j-R})$. Hence, for every interval $I \subset (0, 1)$,

$$\mathbb{P}(Y_j \in I) \leq C(|I| + \varphi^{-2R}) + C\rho^{j-R}.$$

Taking $R = \lfloor j/2 \rfloor$ gives $C|I| + Ce^{-cj}$. If $r \neq 0$, the event in (21) is $|s - rY_j| < \eta$ and is contained in an interval of length $2\eta/|r|$. If $r = 0$, it is empty. $\square$

The second ingredient is a complete-cylinder oscillatory estimate. Its formulation makes the measurability needed for integration by parts explicit.

Lemma 3.4 (Descendant-cylinder estimate). *Let $\{I_w\}$ be cylinders of a fixed depth d . On I_w let $Q_w \in \mathbb{Z} \setminus \{0\}$ be fixed. Let $k > d$ and let A be constant on every depth- k cylinder, with $|A| \leq 1$. Suppose that, below I_w , the support of A is a union of depth- k descendants satisfying $q_k \leq R_w$, where*

$$R_w^2 \leq \varepsilon n |Q_w|.$$

Then

$$(22) \quad \left| \sum_w \int_{I_w} e^{2\pi i n Q_w \alpha} A(\alpha) d\alpha \right| \leq C\varepsilon.$$

Conversely, if a depth- k cylinder J satisfies $q_k^2 \geq \varepsilon^{-1} n |Q|$, then

$$(23) \quad \sup_{\alpha, \alpha' \in J} |nQ(\alpha - \alpha')| \leq C\varepsilon.$$

Proof. Let $q(w)$ be the last denominator of w . If (Q, Q') is the denominator pair of a continuation, the final denominator is $q_k = q(w)Q + q'(w)Q' \geq q(w)Q$. The number of descendants with $q_k \leq R_w$ is therefore $O(R_w^2/q(w)^2)$. On each complete descendant cylinder J ,

$$\left| \int_J e^{2\pi i n Q_w \alpha} d\alpha \right| \leq (\pi n |Q_w|)^{-1}.$$

Thus the contribution below w is at most $C\varepsilon/q(w)^2$. By (14), $q(w)^{-2} \leq 2|I_w|$, and summation proves (22). The cylinder length bound $|J| \leq q_k^{-2}$ gives (23). $\square$

4. MARKED FACTORIZATION AND RESONANCES

For $D > 0$ let $\mathcal{P}_D(L)$ consist of functions

$$(24) \quad F(a, \theta) = \sum_{|v| \leq L^D} c(a, v) e^{2\pi i v \theta}, \quad c(a, v) = 0 \quad (a > L^D),$$

such that $\sum_{a,v} |c(a, v)| \leq L^D$. For times $j_1 < \dots < j_r$ in $\mathcal{J}_n$, call the tuple good if

$$(25) \quad j_{\ell+1} - j_\ell \geq 200H$$

and, for every $i < \ell$,

$$(26) \quad \left| j_\ell - \frac{m_n + j_i}{2} \right| \geq 200H.$$

The number of non-good r -tuples is $O_r(L^{r-1}H)$.

Proposition 4.1 (Marked factorization). *Fix r, D, A . If $(j_1, \dots, j_r)$ is good and $F_1, \dots, F_r \in \mathcal{P}_D(L)$, then*

$$(27) \quad \begin{aligned} & \int_0^1 \prod_{\ell=1}^r F_\ell(a_{j_{\ell+1}}, \theta_{j_\ell}) d\alpha \\ &= \prod_{\ell=1}^r \int_0^1 \int_0^1 F_\ell(a_1(x), \theta) d\theta d\nu(x) + O_{r,D,A}(L^{-A}). \end{aligned}$$

Proof. Expand both in the digit values and in the Fourier modes $v_1, \dots, v_r$, and pull out the scalar coefficient of each digit–Fourier term. The remaining amplitude is a product of cylinder indicators and hence has modulus at most one; the total ℓ^1 -mass of the extracted coefficients is at most L^{Dr} . If every $v_\ell = 0$, the integrand is a product of digit functions. Repeated use of Lemma 3.2 and (25) factors its mean with error $L^{O_{r,D}(1)} \rho^{200H} = O_A(L^{-A})$.

Suppose now that $v_s \neq 0$ and $v_\ell = 0$ for $\ell > s$. By (8), the phase is

$$e^{2\pi i n Q \alpha}, \quad Q = \sum_{\ell=1}^s v_\ell (-1)^{j_\ell} q_{j_\ell}.$$

The continuants grow at least at the Fibonacci rate. The separation condition and $|v_\ell| \leq L^D$ therefore give, deterministically,

$$(28) \quad \frac{1}{2} q_{j_s} \leq |Q| \leq 2L^D q_{j_s}$$

for all sufficiently large n .

The nominal resonance time is

$$R_s = \frac{m_n + j_s}{2}.$$

Put $k_\pm = \lfloor R_s \pm 40H \rfloor$. We now make only local complete-cylinder cuts. At depth j_s retain the cylinder union on which

$$e^{\lambda j_s - \delta H} \leq q_{j_s} \leq e^{\lambda j_s + \delta H}.$$

At depths k_- and k_+ retain, respectively,

$$q_{k_-} \leq e^{\lambda k_- + \delta H}, \quad q_{k_+} \geq e^{\lambda k_+ - \delta H},$$

where $\delta > 0$ is sufficiently small. By (20), the total discarded mass is $O(e^{-cL^{1/2}})$. On every retained prefix and descendant,

$$(29) \quad q_{k_-}^2 \leq e^{-cH} n |Q|, \quad q_{k_+}^2 \geq e^{cH} n |Q|.$$

The factors after s are zero torus modes and hence digit functions. By (26), each lies either at least $100H$ before k_- or at least $100H$ after k_+ . Partition first into complete depth- k_+ cylinders. On each retained cylinder the phase is constant up to $O(e^{-cH})$ by (23). The conditional density of the future has uniformly bounded variation, so Lemma 3.2 replaces all post-resonance digit

factors by their stationary means, with total error $L^{O(1)}(e^{-cH} + \rho^{cH})$ after summation by cylinder mass.

After that replacement, extend the stationary-mean replacement to the discarded depth- k_+ cylinders and restore them. Their total mass is $O(e^{-cL^{1/2}})$, so this costs at most $L^{O_{r,D}(1)}e^{-cL^{1/2}}$. The depth- k_+ cutoff indicator has now disappeared, and the remaining amplitude is measurable on complete depth- k_- descendants. On each depth- $(j_s + 1)$ prefix the integer Q is fixed. The first inequality in (29) and Lemma 3.4 give a contribution $O(e^{-cH})$. Restoring the discarded cylinder unions, and summing all polynomially many Fourier terms, gives the uniform bound

$$(30) \quad \delta_n := L^{O_{r,D}(1)}(e^{-cL^{1/2}} + e^{-cH} + \rho^{cH}) = O_A(L^{-A}).$$

Thus every nonzero Fourier term vanishes to the required order, while the zero terms give the product on the right of (27). $\square$

We also need fixed-radius blocks. The recurrence (9) implies by induction that every torus character in a radius- R window can be reduced, on its digit cylinder, to a character of the two central coordinates. More precisely, for fixed integers $c_{-R}, \dots, c_R$ there are integers A_w, B_w , depending only on the radius- R digit word w , such that

$$(31) \quad \sum_{|t| \leq R} c_t \theta_{j+t} \equiv A_w \theta_j + B_w \theta_{j-1} \pmod{1}.$$

This follows by repeatedly substituting $\theta_{i+1} = \theta_{i-1} - a_{i+1}\theta_i$. On each fixed finite collection of radius- R cylinder words, A_w, B_w range over a finite set. In view of (8), the resulting phase is $n(-1)^j(A_w q_j - B_w q_{j-1})\alpha$.

Let R, K be fixed and call a cylinder-torus monomial a function of the form

$$(32) \quad U_j(\alpha) = u(a_{j-R+1}, \dots, a_{j+R})e^{2\pi i(r\theta_{j-1} + s\theta_j)},$$

where u is supported on finitely many words and $|r| + |s| \leq K$. Its frequency is

$$(33) \quad (-1)^j Q_j(r, s), \quad Q_j(r, s) = sq_j - rq_{j-1}.$$

Proposition 4.2 (Two-block factorization). *For two fixed finite linear combinations U, V of monomials (32), all but $O_{U,V}(LH)$ pairs $j < k$ in $\mathcal{J}_n$ satisfy*

$$(34) \quad \left| \int_0^1 U_j V_k d\alpha - \int U d\hat{\mu}_0 \int V d\hat{\mu}_0 \right| \leq C_{U,V}(e^{-cL^{1/2}} + e^{-cH} + \rho^{cH}),$$

where $\hat{\mu}_0$ is the stationary Gauss natural-extension measure times Haar measure on $\mathbb{T}^2$.

Proof. It suffices to treat monomials. If both torus modes vanish, ordinary two-sided Gauss mixing proves (34) unless the blocks overlap; there are $O(LH)$ overlapping or H -near pairs.

Assume the later mode (r_2, s_2) is nonzero. By Lemma 3.3, outside a complete depth- k cylinder union of mass $O(e^{-cH})$,

$$|Q_k(r_2, s_2)| \geq e^{-cH} q_k.$$

If $k - j \geq CH$, Fibonacci growth makes the later frequency dominate the earlier one; after increasing $C = C(U, V)$,

$$|Q| := |(-1)^j Q_j(r_1, s_1) + (-1)^k Q_k(r_2, s_2)| \geq \frac{1}{2} e^{-cH} q_k.$$

On each depth- $(k + R)$ cylinder, the integer Q and the complete amplitude are fixed. Put

$$t_- := \left\lfloor \frac{m_n + k}{2} - 40H \right\rfloor.$$

Retain the complete depth- k cylinders on which $q_k \geq e^{\lambda k - \delta H}$ and, below each such cylinder, the complete depth- t_- descendants on which $q_{t_-} \leq e^{\lambda t_- + \delta H}$. By (20), the discarded union has mass $O(e^{-cL^{1/2}})$. Since $2\lambda t_- = L + \lambda k - 80\lambda H + O(1)$, every retained descendant satisfies

$$q_{t_-}^2 \leq e^{-cH} n|Q|.$$

Lemma 3.4, with prefix depth $k + R$ and descendant depth t_- , bounds the retained integral by $O(e^{-cH})$. The summation is by complete prefix-cylinder mass, so no cylinder-count factor

is lost. Adding the anti-concentration and denominator-deviation unions gives the right side of (34). The stationary product is zero because the later Haar character is nontrivial.

It remains to suppose that the earlier mode is nonzero and the later mode is zero. Write $Q_j = Q_j(r_1, s_1)$ and

$$t_0 := \frac{m_n + j}{2}, \quad t_- := \lfloor t_0 - 40H \rfloor, \quad t_+ := \lfloor t_0 + 40H \rfloor.$$

Lemma 3.3 removes a complete depth- j cylinder union of mass $O(e^{-cH})$ and leaves $|Q_j| \geq e^{-cH} q_j$. Declare $|k - t_0| \leq 100H$ exceptional. This gives $O(H)$ values of k per j , hence $O(LH)$ pairs in total.

We also retain throughout the complete depth- j cylinder union on which

$$e^{\lambda j - \delta H} \leq q_j \leq e^{\lambda j + \delta H}.$$

By (20), its complement has mass $O(e^{-cL^{1/2}})$. On the retained cylinders,

$$e^{-cH} q_j \leq |Q_j| \leq C_{r_1, s_1} q_j.$$

If $k < t_0 - 100H$, the whole two-block amplitude is measurable before depth t_- . Under the retained depth- j prefixes, keep the complete depth- t_- descendants with $q_{t_-} \leq e^{\lambda t_- + \delta H}$. The complement has mass $O(e^{-cL^{1/2}})$, while on the retained descendants

$$q_{t_-}^2 \leq e^{-cH} n |Q_j|.$$

Lemma 3.4 removes the phase with error $O(e^{-cH})$.

If $k > t_0 + 100H$, also retain the complete depth- t_+ cylinders satisfying $q_{t_+} \geq e^{\lambda t_+ - \delta H}$. Their complement again has mass $O(e^{-cL^{1/2}})$ and, on every retained cylinder,

$$q_{t_+}^2 \geq e^{cH} n |Q_j|.$$

Thus (23) makes the earlier phase constant up to $O(e^{-cH})$ on each retained depth- t_+ cylinder. The later zero-mode block starts at least cH after t_+ , so conditional Gauss mixing, applied on each cylinder and summed by cylinder mass, replaces that block by its stationary mean with error $O(\rho^{cH})$. Extend this stationary-mean replacement to the discarded depth- t_+ cylinders and restore them, at cost $O(e^{-cL^{1/2}})$; no future cutoff indicator then remains in the earlier integral. Finally retain the depth- t_- descendants as above and apply Lemma 3.4 to the remaining earlier phase.

The stationary product is zero because the earlier Haar character is nontrivial. Combining the complete-cylinder errors proves (34); in particular, the denominator-deviation contribution is $O(e^{-cL^{1/2}})$ rather than being hidden in the anti-concentration error. $\square$

5. THE PRINCIPAL POISSON AND STABLE LIMITS

Let

$$Z_{n,j} = a_{j+1} W(\theta_j).$$

If A has the stationary Gauss digit law and Θ is independent and uniform on $\mathbb{T}$, then

$$(35) \quad \mathbb{P}(AW(\Theta) > t) \sim \frac{c_0}{t}, \quad c_0 = \frac{1}{12 \log 2}.$$

Indeed,

$$\nu(A \geq m) = \frac{1}{\log 2} \log \left(1 + \frac{1}{m} \right) \sim \frac{1}{m \log 2},$$

and dominated convergence gives

$$\lim_{t \rightarrow \infty} t \mathbb{P}(AW(\Theta) > t) = \frac{1}{\log 2} \int_0^1 W(\theta) d\theta = \frac{1}{12 \log 2}.$$

$$(36) \quad \Xi_n := \sum_{j \in \mathcal{J}_n} \delta_{(-1)^j Z_{n,j}/L}.$$

Proposition 5.1 (Poisson point process). *In the space of Radon point measures on $\mathbb{R} \setminus \{0\}$ equipped with the vague topology,*

$$(37) \quad \Xi_n \implies \text{PRM}(\Lambda), \quad d\Lambda(x) = \frac{1}{2\pi^2} |x|^{-2} dx.$$

Proof. We verify factorial moment measures against $\varphi_1, \dots, \varphi_r \in C_c^1(\mathbb{R} \setminus \{0\})$. Writing $X_{n,j} = (-1)^j Z_{n,j}/L$, the relevant ordered factorial sum is

$$(38) \quad \sum_{\substack{(j_1, \dots, j_r) \in \mathcal{J}_n^r \\ j_1, \dots, j_r \text{ pairwise distinct}}} \mathbb{E} \prod_{\ell=1}^r \varphi_\ell(X_{n,j_\ell}).$$

Let $c > 0$ be smaller than the distance of all supports from zero. If $\varphi_\ell((-1)^j a W(\theta)/L) \neq 0$, then $a \geq 8cL$.

First cut at $a \leq L^D$, where $D > 2$. By (15), the total contribution of r -tuples for which one digit exceeds L^D and all the others exceed $8cL$ is

$$O\left(L^r \frac{1}{L^D L^{r-1}}\right) = O(L^{1-D}).$$

For $a \leq L^D$, the C^1 function $f_{a,j}(\theta) = \varphi_\ell((-1)^j a W(\theta)/L)$ satisfies $\|f'_{a,j}\|_\infty \leq C_\varphi a/L \leq C_\varphi L^{D-1}$. Jackson's theorem [8, Chap. X], with degree $M = L^{A+r+D+4}$, gives a trigonometric polynomial whose uniform error is $O(L^{-A-r-4})$. Its coefficient ℓ^1 norm is at most $(2M+1)(\|\varphi_\ell\|_\infty + 1)$ and hence is polynomial in L . Increasing the complexity exponent in Proposition 4.1, we may apply that proposition; after summation over $O(L^r)$ tuples, the approximation error is $O(L^{-A-4})$.

Non-good time tuples are $O_r(L^{r-1}H)$ in number. On each of them the probability that all factors are nonzero is $O_r(L^{-r})$ by (15); their total contribution is $O(H/L) = o(1)$. On good tuples, Proposition 4.1 factors the expectation, and its error remains $o(1)$ after summing $O(L^r)$ tuples. Equation (35) gives the vague convergence

$$L\mathbb{P}(AW(\Theta)/L \in dx) \implies c_0 x^{-2} dx \quad (x > 0).$$

To evaluate (38), partition the ordered distinct tuples according to the unique permutation $\sigma \in S_r$ for which $j_{\sigma(1)} < \dots < j_{\sigma(r)}$, and then according to the parity vector $\varepsilon_\ell = (-1)^{j_\ell} \in \{-1, 1\}$. For every fixed σ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_r)$,

$$(39) \quad \#\{(j_1, \dots, j_r) : j_{\sigma(1)} < \dots < j_{\sigma(r)}, (-1)^{j_\ell} = \varepsilon_\ell\} = \frac{1 + o(1)}{r!} \left(\frac{L}{2\lambda}\right)^r.$$

Indeed, the parity-restricted lattice discrepancy in the actual bulk interval is $O(L^{r-1})$, while replacing its length by L/λ contributes $O(HL^{r-1}) = o(L^r)$. Put

$$I_\ell^\varepsilon = \int_0^\infty \varphi_\ell(\varepsilon x) x^{-2} dx.$$

For a fixed ordering and parity vector, factorization and (35) show that the contribution tends to

$$\frac{1}{r!} \left(\frac{c_0}{2\lambda}\right)^r \prod_{\ell=1}^r I_\ell^{\varepsilon_\ell}.$$

Summing first over the $r!$ temporal orderings and then over all parity vectors yields

$$\prod_{\ell=1}^r \left\{ \frac{c_0}{2\lambda} \int_{\mathbb{R}} \varphi_\ell(x) |x|^{-2} dx \right\} = \prod_{\ell=1}^r \int \varphi_\ell d\Lambda, \quad \frac{c_0}{2\lambda} = \frac{1}{2\pi^2}.$$

Thus the ordered factorial moment measures converge to those of the Poisson random measure in (37). If $K \Subset \mathbb{R} \setminus \{0\}$ and $N_n(K)$ denotes the number of displayed points in K , the same iterated conditional tail bound gives, for every $r \geq 1$,

$$\sup_n \mathbb{E}(N_n(K))_r \leq C_K^r,$$

where $(x)_r = x(x-1)\cdots(x-r+1)$. In particular all local factorial moments are uniformly bounded, while the first moment supplies local tightness. Moreover, for $0 \leq u \leq 1$,

$$\sup_n \mathbb{E}(1+u)^{N_n(K)} = \sup_n \sum_{r \geq 0} \frac{u^r}{r!} \mathbb{E}(N_n(K))_r \leq \exp(C_K u).$$

Thus the local counts have uniformly bounded exponential moments. On finitely many disjoint compact continuity sets, the preceding calculation gives convergence of all mixed factorial moments to those of independent Poisson variables, and the exponential bound makes those moments distribution determining. Local tightness and an exhaustion by compact continuity sets now identify every subsequential limit as the Poisson random measure with intensity Λ ; this is the standard factorial-moment criterion [7, Chap. 4]. $\square$

We next control the compensated small jumps. Choose $\chi \in C^1([0, \infty))$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[0, 1]$, and $\chi = 0$ on $[2, \infty)$. For $0 < \varepsilon < 1$, set

$$(40) \quad U_{n,j}^{(\varepsilon)} = (-1)^j Z_{n,j} \chi\left(\frac{Z_{n,j}}{\varepsilon L}\right).$$

Lemma 5.2 (Small jumps). *For a constant C independent of ε ,*

$$(41) \quad \limsup_{n \rightarrow \infty} \frac{1}{L^2} \text{Var} \left(\sum_{j \in \mathcal{J}_n} U_{n,j}^{(\varepsilon)} \right) \leq C\varepsilon.$$

Proof. Since $W \leq 1/8$, Lemma 3.1(ii) gives, uniformly in j, n ,

$$\mathbb{P}(Z_{n,j} > t) \leq \frac{C}{1+t}.$$

For $j \neq k$, (15) similarly gives

$$\mathbb{P}(Z_{n,j} > s, Z_{n,k} > t) \leq \frac{C}{(1+s)(1+t)}.$$

Layer-cake integration, with $M = 2\varepsilon L$, yields

$$(42) \quad \mathbb{E}|U_{n,j}^{(\varepsilon)}|^2 \leq C\varepsilon L, \quad \mathbb{E}|U_{n,j}^{(\varepsilon)} U_{n,k}^{(\varepsilon)}| \leq C \log^2(2+L).$$

The diagonal contribution is $O(\varepsilon L^2)$. Overlapping, H -near, or resonance-near pairs number $O(LH)$, and their total contribution is $O(LH \log^2 L) = o(L^2)$.

For the remaining pairs, cut each digit at $a \leq A_L = L^D$. Since $|U_{n,j}^{(\varepsilon)}| \leq 2\varepsilon L$,

$$\|U_{n,j}^{(\varepsilon)} \mathbf{1}_{\{a_{j+1} > A_L\}}\|_2 \leq C\varepsilon L A_L^{-1/2}.$$

After centering, $\|X - \mathbb{E}X\|_2 \leq 2\|X\|_2$; hence, after division by L , Minkowski's inequality shows that the sum of these tails is $O(\varepsilon L^{1-D/2}) = o(1)$ if $D > 2$. The map $z \mapsto z\chi(z/(\varepsilon L))$ has uniformly bounded Lipschitz norm on $[0, \infty)$, and W is $1/2$ -Lipschitz on $\mathbb{T}$. Hence, on $a \leq A_L$, the truncated summand has Lipschitz norm at most $Ca \leq CL^D$, uniformly in $0 < \varepsilon < 1$. Jackson approximation for Lipschitz functions, with uniform error L^{-A-4} and polynomial degree and coefficient norm, applies as in the proof of Proposition 5.1. Centering doubles the approximation error but does not change its order. Proposition 4.1, with A chosen so large that $L^2 O_A(L^{-A}) = o(1)$, makes the sum of all remaining covariances $o(L^2)$. This proves (41). $\square$

Corollary 5.3 (Principal Cauchy law). *There are deterministic numbers $b_n^{(0)}$ such that*

$$(43) \quad \frac{1}{L} \sum_{j \in \mathcal{J}_n} (-1)^j Z_{n,j} - b_n^{(0)} \implies \text{Cauchy}\left(0, \frac{1}{2\pi}\right).$$

Proof. For fixed ε , split $x = x\chi(|x|/\varepsilon) + x(1 - \chi(|x|/\varepsilon))$. Proposition 5.1 and the continuous mapping theorem give convergence of the second, large-jump part; the digit-tail bound gives the uniform finite- n estimate

$$\sup_n \mathbb{E} \Xi_n \{x : |x| > R\} \leq C/R,$$

so the tails $|x| > R$ may be removed before applying continuous mapping. Center the small part by

$$b_{n,\varepsilon} = \frac{1}{L} \sum_{j \in \mathcal{J}_n} \mathbb{E} U_{n,j}^{(\varepsilon)}.$$

Lemma 5.2 shows that its centered contribution becomes negligible as $\varepsilon \downarrow 0$. The transition shell $\varepsilon < |x| < 2\varepsilon$ is included in the same continuous truncation; its limiting compensated variance is $O(\varepsilon)$.

For the smooth cutoff used here, the limiting large-jump characteristic exponent is, first on $|x| \leq R$,

$$\int_{|x| \leq R} (\exp\{itx(1 - \chi(|x|/\varepsilon))\} - 1) d\Lambda(x).$$

Letting $R \rightarrow \infty$ is legitimate because the expected number of points outside $[-R, R]$ is $O(R^{-1})$. Symmetry cancels the imaginary part, and then dominated convergence as $\varepsilon \downarrow 0$ gives

$$\int_{\mathbb{R}} (\cos(tx) - 1) \frac{dx}{2\pi^2 x^2} = -\frac{|t|}{2\pi}.$$

A diagonal choice $\varepsilon_n \downarrow 0$ gives a deterministic sequence $b_n^{(0)} := b_{n,\varepsilon_n}$ for which (43) holds. $\square$

6. THE CARRY GRAPH AND THE BOUNDED REMAINDER

Let $(\widehat{\Omega}, \widehat{\nu}, \sigma)$ be the two-sided natural extension of the Gauss system and write $x(\omega) = [0; a_1(\omega), a_2(\omega), \dots]$. Whenever order, ceiling, or fractional part is used for a torus coordinate, we identify $\mathbb{T}$ with its canonical representatives in $[0, 1)$. Define the torus cocycle

$$(44) \quad \mathcal{S}(\omega, r, s) = (\sigma\omega, s, r - a_1(\omega)s \pmod{1}).$$

The fibre matrix

$$A_a = \begin{pmatrix} 0 & 1 \\ 1 & -a \end{pmatrix}$$

lies in $\mathrm{GL}_2(\mathbb{Z})$; hence $\widehat{\mu}_0 = \widehat{\nu} \otimes m_{\mathbb{T}^2}$ is invariant.

Lemma 6.1 (Endpoint-controlled continuant recurrence). *Let $a_i \geq 1$ and let a nonzero integer sequence satisfy*

$$z_{i+2} = a_{i+1}z_{i+1} + z_i \quad (0 \leq i < m).$$

If

$$\max(|z_0|, |z_1|, |z_m|, |z_{m+1}|) \leq K,$$

then $m \leq C \log(2K)$ for an absolute constant C .

Proof. Consider a maximal block on which consecutive nonzero terms have alternating signs. Write an interior pair as $(z_i, z_{i+1}) = (u, -v)$ after changing the overall sign, where $u, v > 0$. If alternation persists for two further terms, then

$$z_{i+2} = u - a_{i+1}v > 0, \quad z_{i+3} = a_{i+2}z_{i+2} - v < 0.$$

Consequently $0 < z_{i+2} < v$, and the absolute-value pair is replaced by the ordinary Euclidean remainder pair $(v, u - a_{i+1}v)$. Thus every interior transition of the maximal alternating block is a Euclidean remainder step; only its terminal transition can fail the remainder inequality, and that costs one additional step. Lamé's bound for the Euclidean algorithm (whose extremal quotients are all 1) bounds its length by $C \log(2K)$ when both endpoint pairs are bounded by K . A zero term can occur only at the junction of two such intervals and causes no additional run.

Once two adjacent nonzero terms have the same sign, every later adjacent pair has that sign and

$$|z_{i+2}| \geq |z_{i+1}| + |z_i|;$$

hence the forward absolute values grow at least as Fibonacci numbers. The same argument applied to the recurrence solved backwards controls the final same-sign run. The sequence is

therefore the concatenation of at most two Euclidean runs and one Fibonacci-growth run, each of length $O(\log(2K))$. $\square$

Lemma 6.2 (Gauss–torus mixing). *The system $(\widehat{\Omega} \times \mathbb{T}^2, \widehat{\mu}_0, \mathcal{S})$ is mixing.*

Proof. It is enough to use finite digit-cylinder functions times torus characters. For characters $k, \ell \in \mathbb{Z}^2$, the fibre integral in an m -step correlation vanishes unless

$$(45) \quad k + (A_{a_m} \cdots A_{a_1})^T \ell = 0.$$

Indeed, reverse the matrix order forced by the transpose and, for $0 \leq i \leq m$, put

$$w_i = A_{a_{m-i+1}}^T \cdots A_{a_m}^T \ell = (\xi_i, \eta_i)^T, \quad w_0 = \ell.$$

Since $A_a^T = A_a$, for $0 \leq i < m$,

$$w_{i+1} = A_{a_{m-i}}^T w_i = (\eta_i, \xi_i - a_{m-i}\eta_i)^T.$$

Thus $\xi_{i+1} = \eta_i$; on defining $\xi_{m+1} := \eta_m$, we have

$$\xi_{i+2} = \xi_i - a_{m-i}\xi_{i+1} \quad (0 \leq i < m).$$

Consequently $z_i = (-1)^i \xi_i$ satisfies $z_{i+2} = a_{m-i}z_{i+1} + z_i$. Moreover, $w_m = (A_{a_m} \cdots A_{a_1})^T \ell = -k$ under (45); hence (z_0, z_1) is, up to signs, the coordinate pair of ℓ , and (z_m, z_{m+1}) is the coordinate pair of k . Both endpoint pairs are therefore bounded in terms of the fixed vectors k and ℓ , so Lemma 6.1 applies. Thus, for fixed nonzero k, ℓ , the resonance relation is impossible for all sufficiently large m . The zero modes reduce to mixing of the Gauss natural extension. Density of cylinder functions times characters in L^2 completes the proof. $\square$

We need to transfer fixed windows of the actual curve process to this stationary system. The complete quotient is an infinite-future coordinate, so we record the required full-state form rather than silently treating it as a finite digit function.

For a fixed $R \geq 0$, give

$$(46) \quad \mathcal{X}_R = \mathbb{N}^{2R+1} \times [0, 1]^{2R+1} \times \mathbb{T}^{2R+2}$$

the product topology, with $\mathbb{N}$ discrete, and define the actual window

$$(47) \quad \mathcal{W}_{n,j}^{(R)} = ((a_{j+t+1})_{t=-R}^R, (x_{j+t})_{t=-R}^R, (\theta_{j+t})_{t=-R-1}^R) \in \mathcal{X}_R.$$

The stationary window is defined from the two-sided Gauss–torus system in the same way. Both laws are supported on the corresponding orbit-consistent Borel subset of $\mathcal{X}_R$.

Lemma 6.3 (Uniform full-state transfer). *Let $\eta_{n,j}^{(R)}$ be the law of (47) for Lebesgue α , and let μ_R be the stationary window law. If $G : \mathcal{X}_R \rightarrow \mathbb{C}$ is bounded and its discontinuity set has μ_R -measure zero, then*

$$(48) \quad \sup_{j \in \mathcal{J}_n} \left| \int G d\eta_{n,j}^{(R)} - \int G d\mu_R \right| \longrightarrow 0.$$

Proof. First take a finite digit-cylinder function times a torus character, and let d be the largest initial digit depth on which the chosen test depends. For the raw digit block in (47), $d = j + R + 1$; after replacing complete quotients by a fixed number M of future digits below, one may have $d \leq j + R + M$. Partition into the countably many complete depth- d cylinders, grouped by the finitely many local block words occurring in the test. By (31), on each such cylinder the coefficients A_w, B_w , the integer $Q_j = A_w q_j - B_w q_{j-1}$, and the whole block amplitude are fixed. If $A_w = B_w = 0$, uniform Gauss mixing gives the stationary mean. Otherwise Lemma 3.3, with $\eta = e^{-cH}$, removes, after summing over the finite block collection, a complete cylinder union of mass $O(e^{-cH})$ on which local cancellation may occur. On the complement, $|Q_j| \geq e^{-cH} q_j$.

Put $t = \lfloor (m_n + j)/2 - 40H \rfloor$. Because $j \in \mathcal{J}_n$ and $d - j = O_{R,M}(1)$ for each fixed cylinder test, one has $t > d$ for all sufficiently large n , uniformly in j . Retain the coarser complete depth- j cylinder union on which $q_j \geq e^{\lambda_j - \delta H}$ and, below every retained depth- d prefix, the complete

depth- t descendants on which $q_t \leq e^{\lambda t + \delta H}$. The discarded mass is $O(e^{-cL^{1/2}})$ by (20); on every retained descendant,

$$q_t^2 \leq e^{-cH} n |Q_j|.$$

Lemma 3.4, with prefix depth d , therefore makes every nonzero character $O(e^{-cH} + e^{-cL^{1/2}})$. This proves the required one-block convergence uniformly in j . Finite sums give convergence on the cylinder-character algebra.

Continuous functions of the complete quotients are included by replacing, simultaneously for every $|t| \leq R$, the coordinate $x_{j+t} = [0; a_{j+t+1}, a_{j+t+2}, \dots]$ by its first M future digits. The maximum coordinate error is $O_R(F_M^{-2})$; hence, for continuous G , the test-function error is at most $\omega_G(C_R F_M^{-2})$. Before invoking the finite cylinder-character algebra, restrict all digits used in these truncations to $a \leq K$. Lemma 3.1(ii) and a union bound make the discarded mass $O_R(M/K)$, uniformly for the actual and stationary window laws. These truncated tests are covered by the preceding argument with $d \leq j + R + M$. For fixed M, K only finitely many words remain; first let $n \rightarrow \infty$, then $K \rightarrow \infty$, and finally $M \rightarrow \infty$.

The unital self-conjugate algebra generated by finite digit-cylinder indicators, continuous functions of the real coordinates, and torus characters separates points on each compact digit truncation of $\mathcal{X}_R$. Stone–Weierstrass therefore makes it uniformly dense there; the same algebra is convergence determining and its real span is dense in $L^2(\mu_R)$. The family $\eta_{n,j}^{(R)}$ is uniformly tight: real and torus coordinates are compact, and Lemma 3.1(ii) controls the finitely many digit coordinates. If (48) failed, tightness would give a weakly convergent subsequence, say with limit η . Convergence on the determining algebra forces $\eta = \mu_R$, and the Portmanteau theorem applied to the bounded μ_R -almost-everywhere continuous function G then contradicts the assumed failure. $\square$

For $z = (x, r, s)$ and $d \in \mathbb{N}_0$, define

$$(49) \quad \phi_z(d) = \lceil x(d+r) - s \rceil.$$

Equation (10) shows that the actual carries satisfy $d_{j+1} = \phi_{(x_j, \theta_{j-1}, \theta_j)}(d_j)$. If $e = d + r$, the updated real carry obeys

$$(50) \quad xe \leq e' < xe + 1.$$

Combining this with (6), all actual trajectories and all pullback trajectories started from zero remain below an absolute integer D .

Set

$$(51) \quad \mathcal{R} = \left\{ (x, r, s) : x < \frac{1}{4(D+1)}, \quad \frac{1}{2} < s < \frac{3}{4} \right\}.$$

For $0 \leq d \leq D$ and $z \in \mathcal{R}$,

$$-\frac{3}{4} < x(d+r) - s < -\frac{1}{4},$$

so $\phi_z(d) = 0$. The set $\mathcal{R}$ has positive $\hat{\mu}_0$ -measure. By Lemma 6.2, almost every two-sided orbit visits it infinitely often in the past. Start at time $-N$ with carry zero and iterate to time zero. After the most recent reset the answer no longer depends on N , so it stabilizes almost surely to a measurable function d_* satisfying

$$(52) \quad d_*(\mathcal{S}z) = \phi_z(d_*(z)).$$

Thus the stationary carry extension is a measurable invariant graph and is isomorphic to the mixing base-torus system.

For $R \geq 1$, define $d_j^{(R)}$ by setting the carry equal to zero at time $j - R$ and iterating R times. Let

$$p_R = \hat{\mu}_0 \left(\bigcap_{t=0}^{R-1} \mathcal{S}^{-t} \mathcal{R}^c \right).$$

Ergodicity and $\widehat{\mu}_0(\mathcal{R}) > 0$ imply $p_R \rightarrow 0$. Whenever the actual orbit has a reset during $[j-R, j)$, $d_j = d_j^{(R)}$. The no-reset indicator has stationary-null boundary, so Lemma 6.3 gives

$$(53) \quad \sup_{j \in \mathcal{J}_n} \mathbb{P}(d_j \neq d_j^{(R)}) \leq p_R + o_n(1).$$

We now prove the weak law for the remainder in (11).

Proposition 6.4 (Bounded-remainder weak law).

$$(54) \quad \frac{1}{L} \sum_{j \in \mathcal{J}_n} (-1)^j (B_j - \mathbb{E}B_j) \longrightarrow 0 \quad \text{in probability.}$$

Proof. Let $B_j^{(R)}$ be the bounded remainder obtained by replacing d_j with $d_j^{(R)}$. For fixed R , it is a bounded function of a fixed past window, the torus coordinates, and the current real complete quotient. Its discontinuities lie on countably many Gauss boundaries and finitely many ceiling or fractional-part hypersurfaces, all of stationary measure zero.

By the L^2 -density established after Lemma 6.3, choose real-valued cylinder–torus polynomials $P_{R,M}$ that are finite real linear combinations of the monomials in (32), of finite radii $R_M \geq R$ that may grow with M , such that

$$(55) \quad \|B^{(R)} - P_{R,M}\|_{L^2(\mu_{R_M})} =: \delta_{R,M} \longrightarrow 0.$$

They may be taken real-valued because replacing an approximant by its real part does not increase its distance from the real-valued function $B^{(R)}$. For each fixed R, M , apply Lemma 6.3 with window radius R_M to $|B^{(R)} - P_{R,M}|^2$ to obtain

$$(56) \quad \sup_{j \in \mathcal{J}_n} \mathbb{E}|B_j^{(R)} - P_{R,M,j}|^2 = \delta_{R,M}^2 + o_n(1).$$

For fixed R, M , Proposition 4.2 shows that the variance of $\sum_{j \in \mathcal{J}_n} (-1)^j (P_{R,M,j} - \mathbb{E}P_{R,M,j})$ is $o(L^2)$. The diagonal contributes $O(L)$; the $O(LH)$ exceptional pairs contribute $O(LH) = o(L^2)$; and all other pairs contribute $O(L^2\delta_n) = o(1)$, with δ_n as in (30). Lemma 6.3 gives $\sup_j |\mathbb{E}P_{R,M,j} - \int P_{R,M} d\mu_{R_M}| = o(1)$, so replacing the stationary means in Proposition 4.2 by the actual means adds only $o(L^2)$ to the covariance sum.

Write $\tilde{X} = X - \mathbb{E}X$. Minkowski's inequality and (56) give

$$(57) \quad \left\| \frac{1}{L} \sum_{j \in \mathcal{J}_n} (-1)^j (\tilde{B}_j^{(R)} - \tilde{P}_{R,M,j}) \right\|_2 \leq \frac{2}{L} \sum_{j \in \mathcal{J}_n} \|B_j^{(R)} - P_{R,M,j}\|_2 \leq C\delta_{R,M} + o_n(1).$$

First let $n \rightarrow \infty$, then $M \rightarrow \infty$. This proves the analogue of (54) with $B_j^{(R)}$ for every fixed R .

Finally, boundedness and (53) imply

$$\sup_{j \in \mathcal{J}_n} \|B_j - B_j^{(R)}\|_2 \leq C\sqrt{p_R + o_n(1)}.$$

A second centered Minkowski estimate yields

$$(58) \quad \limsup_{n \rightarrow \infty} \left\| \frac{1}{L} \sum_{j \in \mathcal{J}_n} (-1)^j (\tilde{B}_j - \tilde{B}_j^{(R)}) \right\|_2 \leq C\sqrt{p_R}.$$

Letting $R \rightarrow \infty$ completes the proof. The order of limits is $n \rightarrow \infty$, then $M \rightarrow \infty$, then $R \rightarrow \infty$. $\square$

7. STOPPING, CENTERING, AND COMPLETION OF THE PROOF

The continuous height has the exact form

$$(59) \quad C_j = \frac{n}{q_j + q_{j-1}x_j}, \quad N_j = C_j - E_j, \quad 0 \leq E_j \leq E_*.$$

Lemma 7.1 (Stopping time and end terms). *With $H = L^{3/4}$,*

$$(60) \quad \tau_n = \frac{L}{\lambda} + O_{\mathbb{P}}(H).$$

Moreover,

$$(61) \quad \frac{1}{L} \left[S_n(\alpha) - \sum_{j \in \mathcal{J}_n} (-1)^j \Phi(x_j, u_j) \right] \longrightarrow 0 \quad \text{in probability.}$$

Proof. Since $q_j \leq q_j + q_{j-1}x_j < 2q_j$, (59) gives a deterministic bracketing. If $q_j \leq n/(2(E_* + 1))$, then $C_j \geq E_* + 1$ and hence $N_j \geq 1$. If $q_j > n$, then $C_j < 1$ and, because N_j is a nonnegative integer, $N_j = 0$. Thus τ_n lies between the hitting times at which q_j crosses these two fixed multiples of n . Applying (16) at $L/\lambda \pm CH$ to both thresholds, and then choosing C large, proves (60).

Also,

$$(62) \quad |\Phi(x, u)| \leq \frac{1}{8x} + \frac{1}{2} \leq \frac{a_1(x)}{8} + \frac{5}{8}.$$

Lemma 3.1(ii) therefore gives

$$(63) \quad \sup_{j,n} \mathbb{P}(|\Phi(x_j, u_j)| > t) \leq \frac{C}{1+t}.$$

On the event $|\tau_n - m_n| \leq CH$, the indices omitted in (61) lie in a deterministic union of $O(H)$ positions. The probability that one of their costs exceeds εL is $O(H/L) = o(1)$. The expected sum of the remaining costs, truncated at εL , is $O(H \log L)$ by integration of (63); divided by L this tends to zero. Finally let C grow and use (60). $\square$

Proof of Theorem 1.1. By Proposition 2.2, Corollary 5.3, and Proposition 6.4, there are deterministic c_n such that

$$(64) \quad \frac{1}{L} \sum_{j \in \mathcal{J}_n} (-1)^j \Phi(x_j, u_j) - c_n \implies \text{Cauchy}\left(0, \frac{1}{2\pi}\right).$$

Lemma 7.1 and Slutsky's theorem replace the bulk sum by S_n/L .

It remains to remove the deterministic centering. For almost every α ,

$$\{k(1 - \alpha)\} = 1 - \{k\alpha\}, \quad S_n(1 - \alpha) = -S_n(\alpha).$$

Thus $X_n = S_n/\log n$ has an exactly symmetric law. The family $X_n - c_n$ is tight by (64); symmetry makes $X_n + c_n$ tight as well. Since their deterministic center difference is $2c_n$, the sequence (c_n) is bounded. If $c_{n_k} \rightarrow c$, then $X_{n_k} \Rightarrow Z + c$, where $Z \sim \text{Cauchy}(0, 1/(2\pi))$. The limit must be symmetric about zero, and a nondegenerate translated Cauchy law has this symmetry only when $c = 0$. Hence every convergent subsequence of (c_n) tends to zero, so $c_n \rightarrow 0$.

The Cauchy distribution is continuous; therefore convergence in distribution is equivalent to the asserted convergence of distribution functions. $\square$

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